

DATE:

EXPT NO:05

DATA VISUALIZATION

AIM:

To visualize the Earthquake dataset using Python and create different charts to understand earthquake patterns, magnitude distributions, and identify important geographic features.

PROCEDURE:

1. Start the program.
2. Import Pandas, Matplotlib, and Seaborn libraries.
3. Load the Earthquake CSV dataset.
4. Display the first few records using head().
5. Create a bar chart to compare seismic events based on magnitude.
6. Create a pie chart to show the relative magnitude distribution among seismic events.
7. Create a line chart to analyze seismic trends across recorded events.
8. Add suitable titles, axis labels, and legends.
9. Display all the visualizations.
10. Analyze the graphs and identify important patterns.
11. Stop the program.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("earthquake_data.csv")

# Display dataset
print(df.head())

# Select numerical columns
num_cols = df.select_dtypes(include="number").columns
print("Numerical columns:", num_cols.tolist())

# 1. BAR CHART
plt.figure(figsize=(8, 5))
df[num_cols[0]].head(10).plot(kind="bar")
plt.title("earthquake_data - Bar Chart")
plt.xlabel("Earthquake Event")
plt.ylabel(num_cols[0])
plt.show()
```

```

# 2. LINE CHART
plt.figure(figsize=(8, 5))
plt.plot(df[num_cols[0]].head(20), marker="o")
plt.title("earthquake_data - Line Chart")
plt.xlabel("Earthquake Event")
plt.ylabel(num_cols[0])
plt.grid()
plt.show()

# 3. PIE CHART
plt.figure(figsize=(7, 7))
df[num_cols[0]].head(10).plot(kind="pie", autopct="%1.1f%%")
plt.title("earthquake_data - Pie Chart")
plt.ylabel("")
plt.show()

# 4. BOX PLOT
plt.figure(figsize=(8, 5))
sns.boxplot(y=df[num_cols[0]])
plt.title("earthquake_data - Box Plot")
plt.ylabel(num_cols[0])
plt.show()

# 5. HISTOGRAM
plt.figure(figsize=(8, 5))
plt.hist(df[num_cols[0]], bins=10)
plt.title("earthquake_data - Histogram")
plt.xlabel(num_cols[0])
plt.ylabel("Frequency")
plt.show()

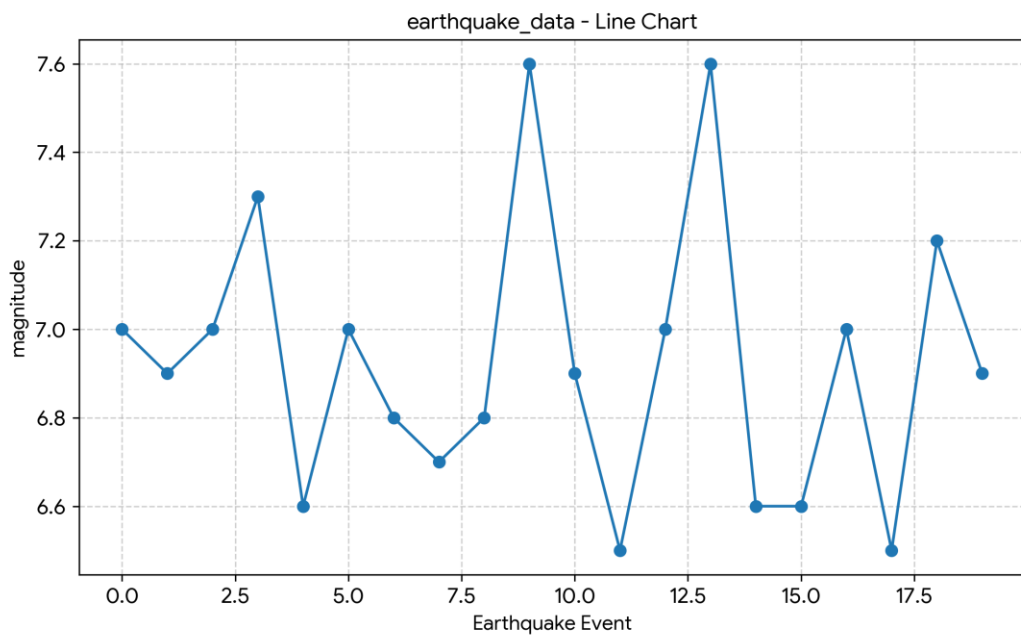
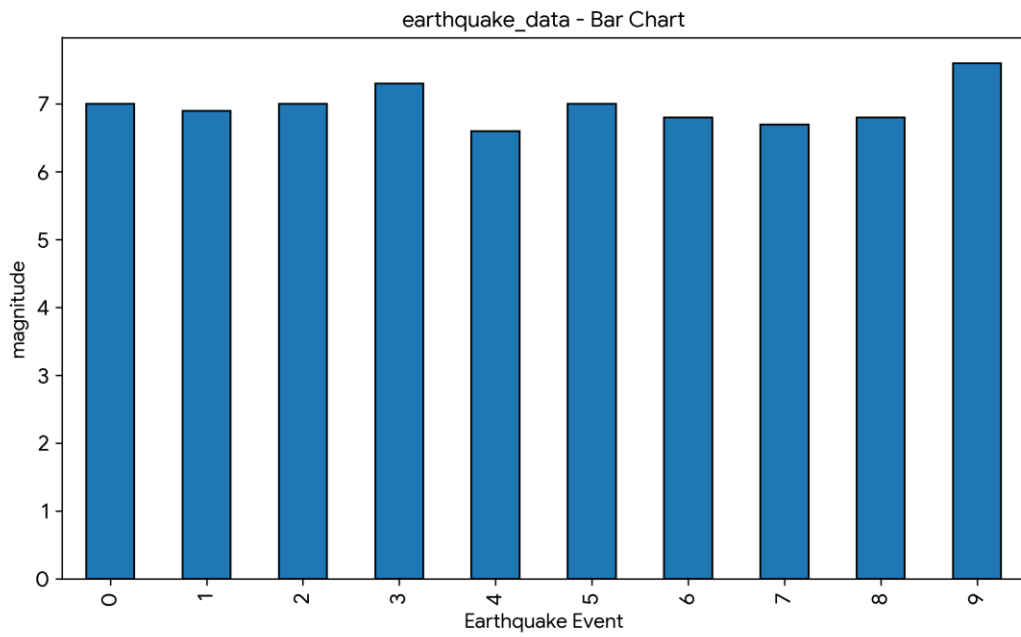
```

OUTPUT:

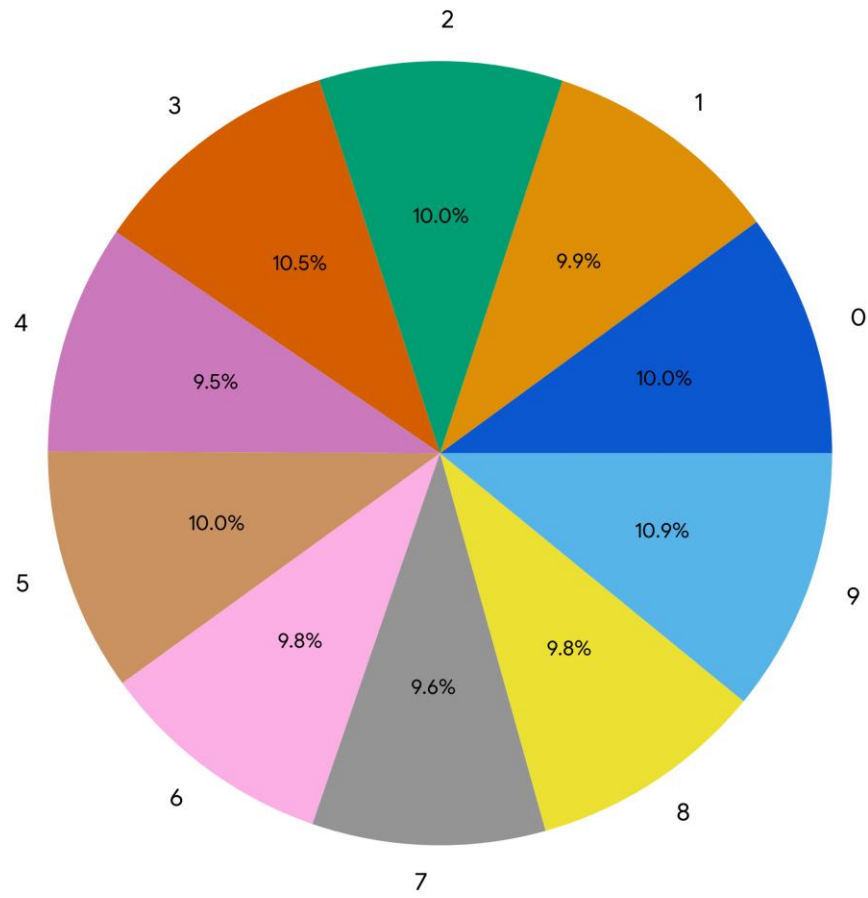
First 5 Records:

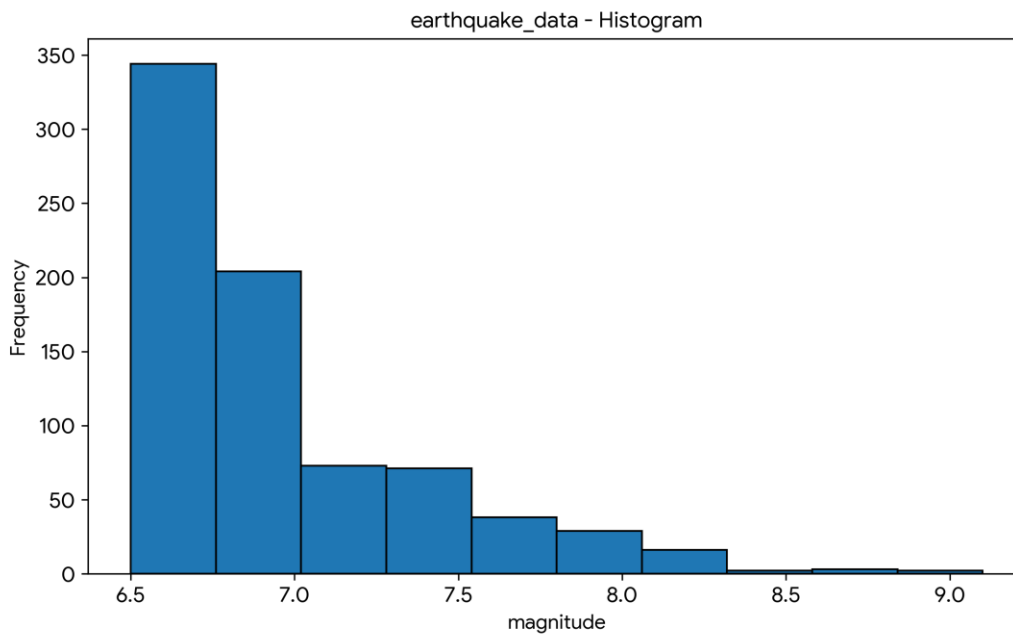
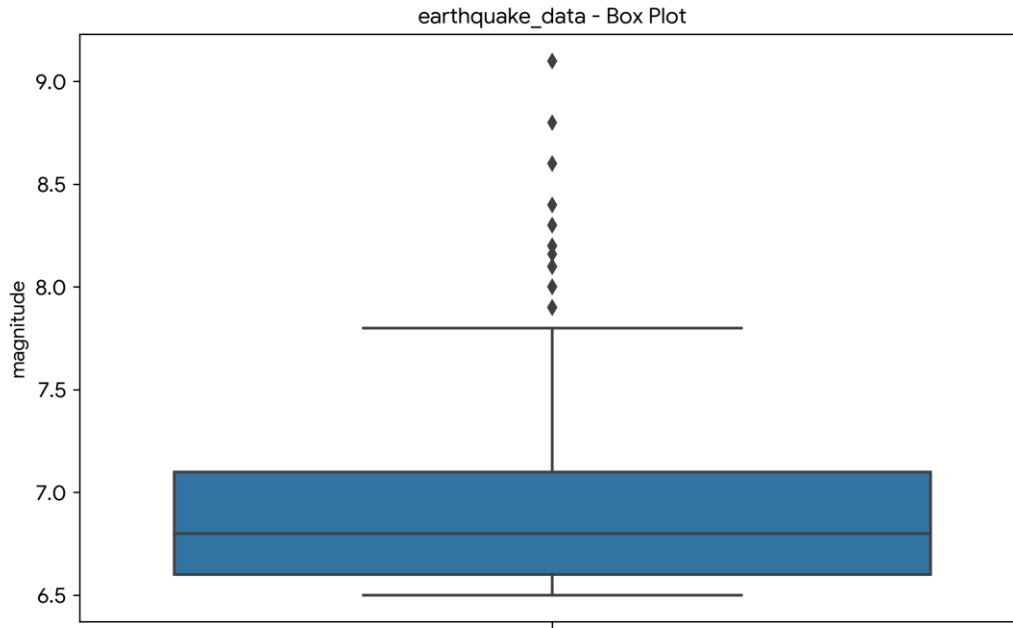
	mmi	alert	tsunami	sig	net	nst	dmin	gap	magType	depth	latitude	longitude	location	continent	country	title	magnitude	date_time	cdi
0	M 7.0	-	18 km SW of Malango, Solomon Islands							7.0	22-11-2022 02:03								8
7	green		1	768	us	117	0.509	17.0	mww	14.000	-9.7963								159.596
													Malango, Solomon Islands	Oceania	Solomon Islands				
1	M 6.9	-	204 km SW of Bengkulu, Indonesia							6.9	18-11-2022 13:37								4
4	green		0	735	us	99	2.229	34.0	mww	25.000	-4.9559								100.738
													Bengkulu, Indonesia	NaN		NaN			
2										7.0	12-11-2022 07:09								3
3	green		1	755	us	147	3.125	18.0	mww	579.000	-20.0508								-178.346
													NaN	Oceania		Fiji			
3	M 7.3	-	205 km ESE of Neiafu, Tonga							7.3	11-11-2022 10:48								5
5	green		1	833	us	149	1.865	21.0	mww	37.000	-19.2918								-172.129
													Neiafu, Tonga	NaN		NaN			
4										6.6	09-11-2022 10:14								0
2	green		1	670	us	131	4.998	27.0	mww	624.464	-25.5948								178.278
													NaN			NaN			

Numerical columns: ['magnitude', 'cdi', 'mmi', 'tsunami', 'sig', 'nst', 'dmin', 'gap', 'depth', 'latitude', 'longitude']



earthquake_data - Pie Chart





RESULT:

The Earthquake dataset was successfully visualized using bar chart, line chart, pie chart, box plot, and histogram. The visualizations helped identify data distribution, trends, variations, and important patterns in the dataset.